What is React?

React is a JavaScript library created by Facebook. It serves as a User Interface (UI) library and is a tool for building UI components.

Adding React to an HTML Page

This quickstart tutorial demonstrates how to add React to a page. Here's an example:

Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Test React</title>
  <!-- Load React API -->
  <script src="https://unpkg.com/react@16/umd/react.production.min.js"></script>
  <!-- Load React DOM -->
  <script src="https://unpkg.com/react-dom@16/umd/react-dom.production.min.js"></script>
  <!-- Load Babel Compiler -->
  <script src="https://unpkg.com/babel-standalone@6.15.0/babel.min.js"></script>
</head>
<body>
  <script type="text/babel">
    // JSX Babel code goes here
  </script>
</body>
</html>
```

What is Babel?

Babel is a JavaScript compiler that translates markup and programming languages into JavaScript. With Babel, developers can use the latest JavaScript features, such as ES6 (ECMAScript 2015). It is particularly useful in React for converting JSX into JavaScript. Note that `<script type="text/babel">` is necessary for using Babel.

What is JSX?

JSX stands for JavaScript XML. It is an XML/HTML-like extension to JavaScript. For example:

```
const element = <h1>Hello World!</h1>;
```

As shown above, JSX is neither pure JavaScript nor HTML. It is an XML syntax extension to JavaScript that leverages the full power of ES6. JSX tags can have tag names, attributes, and children, similar to HTML. Attributes wrapped in curly braces `{}` contain JavaScript expressions, and unlike HTML, JSX does not use quotes around attribute values.

React DOM Rendering

The method `ReactDOM.render()` is used to display HTML elements:

Example:

```
<div id="id01">Hello World!</div>

<script type="text/babel">
ReactDOM.render(
  <h1>Hello React!</h1>,
  document.getElementById('id01'));
</script>
```

JSX Expressions

Expressions in JSX are wrapped in curly braces `{}`.

Example:

```
<div id="id01">Hello World!</div>

<script type="text/babel">
const name = 'John Doe';
ReactDOM.render(
  <h1>Hello {name}!</h1>,
  document.getElementById('id01'));
</script>
```

React Elements

React applications are often structured around a single HTML element, frequently referred to as the root node.

Example:

```
const element = <h1>Hello React!</h1>;
ReactDOM.render(element, document.getElementById('root'));
```

React elements are immutable. Once created, they cannot be changed; the only way to update the UI is to render a new element:

Example:

```
function tick() {
    const element = <h1>{new Date().toLocaleTimeString()}</h1>;
    ReactDOM.render(element, document.getElementById('root'));
}
setInterval(tick, 1000);
```

React Components

React components can be declared either as JavaScript functions or as ES6 classes.

Function Component Example:

```
function Welcome() {
    return <h1>Hello React!</h1>;
}
ReactDOM.render(<Welcome />, document.getElementById('root'));
```

Class Component Example:

```
class Welcome extends React.Component {
  render() {
    return <h1>Hello React!</h1>;
  }
}
ReactDOM.render(<Welcome />, document.getElementById('root'));
```

React Component Properties

Components can accept properties, referred to as props.

Function Component with Props Example:

```
function Welcome(props) {
    return <h1>Hello {props.name}!</h1>;
}
ReactDOM.render(<Welcome name="John Doe"/>, document.getElementById('root'));
```

Class Component with Props Example:

```
class Welcome extends React.Component {
  render() {
    return <h1>Hello {this.props.name}</h1>;
  }
}
ReactDOM.render(<Welcome name="John Doe"/>, document.getElementById('root'));
```

JSX Compiler

While these examples compile JSX in the browser, it's recommended to perform compilation separately for production code.

Create React Application

Facebook has developed Create React App, a setup with everything needed to build a React application. It includes a development server, Webpack for compilation, ESLint for code quality, and more.

To start, run the following command in your terminal:

```
npx create-react-app react-tutorial
```

Ensure you have Node.js version 5.2 or higher, or you may need to install npx separately:

```
npm i npx
```

After installation, navigate to your new React project:

```
cd react-tutorial
npm start
```

This begins the development server and sets up your development application.